

Dynamic Edge Server Cache Update via Handoff/Handin

Notation

Symbol	Meaning
N_{prev}	Global static list of EDs (previous value)
N_{curr}	Snapshot of EDs currently in coverage area
X_{prev}	Models cached on Edge Server (previous value)
X_{curr}	Updated Edge Server cache
D	ModelDemand: mapping of EDs to models they demand
$handin$	EDs handed in to the Edge Server
$handoff$	EDs handed off from the Edge Server
C	ModelEDCount: dict of {model : list of EDs demanding it}
M	Utility matrix over models: (#EDs, Storage, Marginal Utility)
X	Models sorted by Model Utility, non-increasing
L	<i>temp_list</i> : models selected to cache this round
S	Total storage capacity of the Edge Server
k, n	Number of models, number of EDs

Algorithm 1: Model Caching Strategy

Input: N_{curr} : snapshot of EDs in coverage area (Global Snapshot class); N_{prev} : previous Global static list of EDs; X_{prev} : Edge Server's previously cached models

Output: X_{curr} : updated Edge Server cache

Snapshot Inputs

1: $N_{curr} \leftarrow \text{GlobalSnapshot.getCurrentEDs}()$

Model Demand

2: **for** each ED $i \in N_{curr}$ **do**
 3: $D[i] \leftarrow i.\text{requestedModel}$
 4: **end for**

Compare Current vs. Previous EDs

5: Compare N_{curr} with N_{prev} using Global Model Caching List and Edge Server dereference to confirm handin/handoff membership

Handoff / Handin Lists

6: $handoff \leftarrow N_{prev} \setminus N_{curr}$
7: $handin \leftarrow N_{curr} \setminus N_{prev}$

Build ModelEDCount

8: $C \leftarrow \{\}$
9: **for** each ED $i \in N_{curr}$ **do**
10: $m \leftarrow D[i]$
11: $C[m].append(i)$
12: **end for**

Build Utility Matrix (greedy metric)

13: **for** each model $m \in C.keys()$ **do**
14: $M[m].NumEDs \leftarrow |C[m]|$ $\triangleright$ (1) # EDs demanding m
15: $M[m].Storage \leftarrow StorageSize(m)$ $\triangleright$ (2) Storage required
16: $M[m].Utility \leftarrow MarginalUtility(m, C[m])$ $\triangleright$ (3) Marginal Utility w.r.t. EDs wanting m
17: **end for**

$\triangleright$ worst case: all EDs map to one model $\Rightarrow O(k \cdot n) \approx O(n)$

Sort Models by Utility

18: $X \leftarrow \text{SORT}(M.keys(), \text{key} = M[m].Utility, \text{order} = \text{DESCENDING})$

Build temp_list via Storage Allocation

19: $L \leftarrow \emptyset, usedStorage \leftarrow 0$
20: **for** each model $m \in X$ **do**
21: **if** $usedStorage + M[m].Storage \leq S$ **then**
22: $L \leftarrow L \cup \{m\}$
23: $usedStorage \leftarrow usedStorage + M[m].Storage$
24: **end if**
25: **end for**

Update Global EDs List

26: $EdgeServer.N \leftarrow N_{curr}$

Update Cache using LRU Strategy

27: $X_{curr} \leftarrow \emptyset$
28: **for** each model $m \in L$ **do**
29: **if** $m \in X_{prev}$ **then**
30: $X_{curr} \leftarrow X_{curr} \cup \{m\}$
31: $LRU.touch(m)$
32: **else**
33: $X_{curr} \leftarrow X_{curr} \cup \{m\}$
34: $LRU.insert(m)$
35: **end if**
36: **end for**

37: $EdgeServer.cachedModels \leftarrow X_{curr}$

38: **return** X_{curr}